

Unit 3: Practical Skills in Chemistry I

IAS compulsory unit

Externally assessed

Unit description

Introduction

This unit consists of a written practical examination, covering the skills and techniques developed during practical work in Units 1 and 2.

Although the unit content contains eight core practical activities, the examination will not be limited to recall of these core practicals but may include questions where students are expected to apply their knowledge to new practical situations.

Students should, therefore, develop their practical skills by completing a range of different practicals that require a variety of different techniques.

As students carry out practical activities, they should be encouraged to write laboratory reports using appropriate scientific, technical and mathematical language, conventions and symbols.

Development of practical skills, knowledge and understanding

Students are expected to develop experimental skills and a knowledge and understanding of the necessary techniques by carrying out a range of practicals while they study Units 1 and 2.

This unit will assess students' knowledge and understanding of the practical procedures and techniques they develop.

To prepare for assessment of this unit, centres should give students opportunities to carry out practical activities, collect and analyse data, and draw conclusions. Students should – at the least – carry out the eight core practicals in class. By completing these practicals students will be able to:

- follow and interpret experimental instructions, covering the full range of laboratory exercises set throughout the course, with minimal help from the teacher
 - always work with interest and enthusiasm in the laboratory, completing most laboratory exercises in the time allocated
 - manipulate apparatus, use chemicals, carry out all common laboratory procedures and use data logging (where appropriate) with the highest level of skill that may be reasonably expected at this level
 - work sensibly and safely in the laboratory, paying due regard to health and safety requirements without the need for reminders from the teacher
 - gain accurate and consistent results in quantitative exercises, make most of the expected observations in qualitative exercises and obtain products in preparations of high yield and purity.
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Assessment information

- First assessment: June 2019.
 - The assessment is 1 hour and 20 minutes.
 - The assessment is out of 50 marks.
 - Students must answer all questions.
 - This paper may include short-open, open-response and calculation questions.
 - This paper will include a minimum of 6 marks that target mathematics at Level 2 or above.
 - Students will be expected to apply their knowledge and understanding of practical skills to familiar and unfamiliar situations.
 - Calculators may be used in the examination (see *Appendix 8: Use of calculators*).
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The question paper will assess the following abilities.

Independent thinking in a practical context

- Solve problems set in a practical context.
- Apply scientific knowledge to practical contexts.

Use and application of scientific methods and practices

- Identify and state how to control variables to improve experimental validity.
- Present data in appropriate ways.
- Evaluate results and draw conclusions.
- Appreciate measurement uncertainties and errors.
- Comment on the method for an experiment.

Numeracy and the application of mathematical concepts in a practical context

- Plot and interpret graphs.
- Process and analyse data using appropriate mathematical skills.
- Use appropriate numbers of significant figures based on the experimental data.
- Consider the accuracy and precision of data.

Use of apparatus and equipment

- Recognise a range of laboratory apparatus and select appropriate apparatus for a particular scenario.
- Understand how to use a range of apparatus and techniques appropriate to the knowledge and understanding included in this specification.
- Consider the range and resolution of apparatus.
- Identify health and safety issues and discuss how these may be dealt with.

Questions may require students to:

- recall and/or interpret observations relating to tests for ions and gases in Units 1 and 2
 - recall and/or interpret observations relating to tests for organic functional groups in Units 1 and 2
 - manipulate data and comment on experimental methods and techniques for a range of experiments involving measurements in Units 1 and 2, including molar mass calculations, titrations, thermochemical investigations and simple kinetics experiments
 - comment on experimental methods and techniques in the preparation of inorganic or organic compounds in Units 1 and 2.
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